

Sophia Schiffer

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5541 S Everett Ave, Apt. 811, Chicago, IL 60637

EDUCATION

Northwestern University

Bachelor of Science in **Mechanical Engineering**

Evanston, Illinois

June 2024

GPA: 3.740/4.000

EMPLOYMENT & PROJECTS

Focal Point, LLC

Mechanical Engineer I – Custom Design

Chicago, Illinois

November 2024 – Present

- Design custom aluminum housings and lenses to match customer-specified patterns for all Focal Point's linear product lines (e.g. Seem 1, Seem 2, Focus, etc. with various suspended or recessed mounting types)
- Compile bills of material for custom lighting orders to incorporate designed components, LEDs and drivers for specified lumen outputs, and special requests for mounting, sensor, optic, and finish offerings

Mechanical Engineering Internship Projects

June 2023 – August 2023

- Developed step light for hospital room meeting even distribution and low 40-lumen output requirements
- Prototyped and developed ceiling cover shrouds for recessed narrow-beam downlights with high tilt angles up to 40°

Interactive & Emergent Autonomy Lab

Active Control for WALTER Soft Robot

Evanston, Illinois

July 2022 – June 2024

- Tuned primitive walking gaits for automated tree-search learning of legged locomotion with 4-minute training time
- Implemented RRT-based path planner for obstacle navigation in simulation

Publication: Ketchum, Jake, Sophia Schiffer, Muchen Sun, Pranav Kaarthik, Ryan Landon Truby and Todd D. Murphey.

"Automated Gait Generation For Walking, Soft Robotic Quadrupeds." *ArXiv* abs/2310.00498 (2023): n. pag.

Griffin Museum of Science and Industry

Guest Engagement Facilitator I

Chicago, Illinois

June 2024 – November 2024

- Provided informal education and demos to museum guests on storms, electricity, and technology
- Lead guided tours of the U-505 (WWII German U-boat) and coal mine, explaining the both history and machinery

Autonomous Pin Shooter Capstone Project

Systems Integrator

Evanston, Illinois

November 2023 – December 2023

- Designed robust AprilTag tracking pipeline for detecting precise placement of Nerf Modulus Blaster
- Implemented MoveIt2 motion planning to grasp and place Nerf guns with Franka Emika Panda robot

SKILLS

- **Programming:** Python, MATLAB, C, C++, ROS/ROS2
- **Engineering Software:** SOLIDWORKS, Fusion360, Ansys (FEA), Arduino, Siemens NX, Oracle
- **Shop:** Band Saw, CNC, Drill Press, Lathe, Mill, 3D Printer, Sand Casting, Sewing Machine
- **Design:** Design for Manufacturing, Injection Molding, Sheet Metal, Design for Aesthetics, LED Lighting
- **Creative Software:** Inkscape, Adobe Photoshop, Digital Audio Workstations (FL Studio)

LEADERSHIP

Northwestern Club Spikeball Marketing Executive, 2023-2024 Wildcat Welcome/McCormick School of Engineering Peer Advisor